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Wireless Communication

Units

1. Wireless channels
2. Cellular Architecture
3. Digital Signaling for fading channels
4. Multipath Mitigation techniques.
5. Multiple Antenna techniques.

* Unit 1

Wireless Channels

Wireless:- Information or signals can be travel without any wire/cable/twisted or coaxial cables/optical fibre etc.

Wireless Communication has much more importance in →

- Mobile App development
- Cyber security
- Network Programming
- IOT
- Data science

* Reference of wireless Communication in Mobile

App development

- 1) Wifi (wireless fidelity)
- 2) Bluetooth
- 3) NFC (Near field communication) (Scan & pay etc)
- 4) Infrared/Zigbee (less range than bluetooth)

Rel

* Modern era & Ancient era communication:-

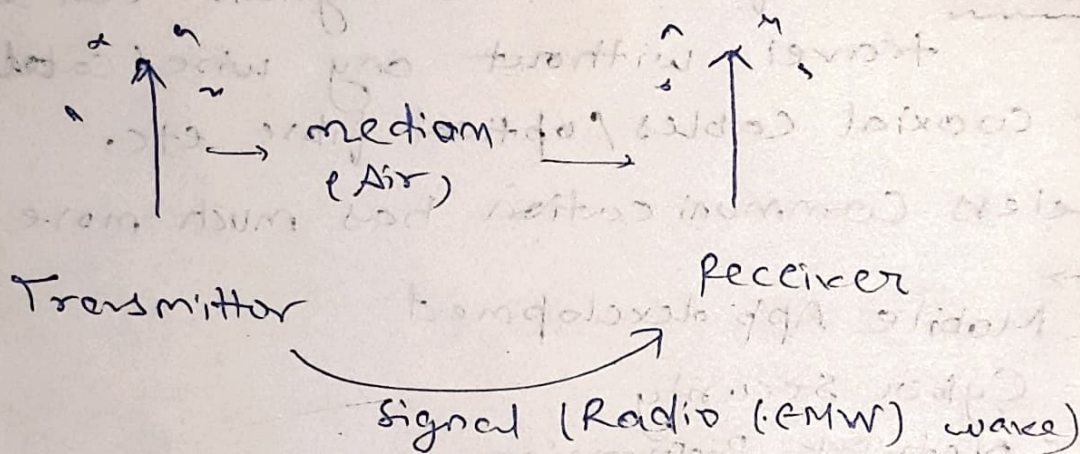
- Provide smoke for giving signals

- Pigeons / Birds carry information from one place to another.

Internet cables can be replaced by Wifi routers

Car Phones can be replaced by Car buds.

* Components of Wireless Communication.



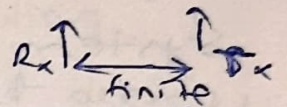
* Basic terms Used in WC:-

1. Propagation
2. Modes of propagation
3. Multipath Propagation
4. Fading
5. Shadowing
6. Path loss

1. Propagation:- It is a tendency of radio waves to travel from one point to another or into various parts of the atmosphere.

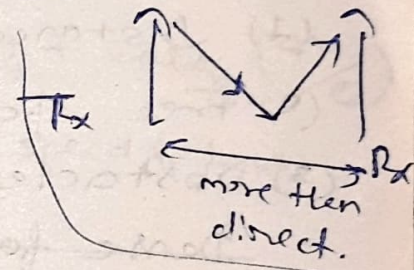
2. Modes of propagation:- The radiated signal from the transmitter reaches to the receiver in 3 different modes.

(a) Direct mode $\Rightarrow$ distance between transmitter & receiver is finite.



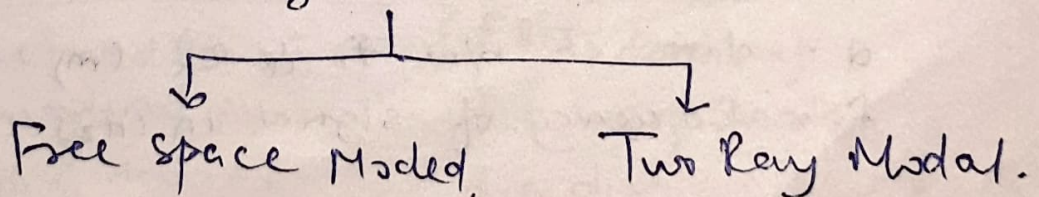
(b) Ground wave propagation $\Rightarrow$ distance between transmitter & receiver, is larger than direct. Quality is differ in this mode.

(c) space (sky) wave propagation
Distance between transmitter & receiver is infinite.



from different layers of atmosphere.
ET = set top box (DTH)

3. Multipath Propagation:- Results in Radio signals reaching the receiving antenna by two/more path causes of multipath include reflection & refraction part from various objects such as mountain, buildings.



* Free Space Path loss :-

(a) Large Scale path loss -

It refers to the reduction in power density of an EM signal as it propagates through space over a long distance in wireless communication system. It is a key component in wireless link budget calculation. & affects signal strength, coverage and system performance.

→ Large scale path loss accounts for signal power attenuation. Due to -

- (1) Distance between the transmitter & Receiver.
- (2) Free space spreading
- (3) Obstacles (Buildings, Mountains, Underground, Dense forest etc.)
- (4) Atmospheric condition.

(b) Path loss models :-

(1) Free space Path loss model (dB)

It is used when there is a clear line of sight (LOS)

$$FSPL (dB) = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55$$

d = distance b/w Tx & Rx (m)

f = frequency of signal in (Hz)

$$FSPL (dB) = 20 \log_{10}(d) + 20 \log_{10}(f) + 32.44$$

if d = in km

if f = in MHz

Free space path loss is the reduction in signal power as a radio wave travels in free space.

An Ideal environment is that when there is no obstacle or reflection or ~~de~~ reflection.

It ~~is~~ describe how signal strenght decreases with distance & frequency is in ideal line of sight (LOS) channel.

$$FSPL (dB) = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10} \left(\frac{4\pi}{c} \right)$$

by solving $20 \log_{10} \left(\frac{4\pi}{c} \right)$ (c is speed of light)

$$FSPL (dB) = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55$$

(In standard units)

* It is used in Wireless Communication Engineering

* ~~link~~ link budget Calculation.

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

Important

P_r = Received Power

P_t = transmitted power

G_t = gain of transmitter

G_r = gain of receiver

It describes received power over a free space channel.

(FRIS Formula)

λ = wavelength.

d = distance b/w P_t & P_r

(5)

$$\left[\text{Path loss} = \left(\frac{4\pi d}{\lambda} \right)^2 \right] \left\{ \begin{array}{l} \lambda = c/f \\ c = 3 \times 10^8 \text{ m/s} \\ f = \text{frequency} \end{array} \right\}$$

$$\text{FSPL (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55$$

Energy is spread over a spherical Area

$$A = 4\pi d^2$$

If the Transmitted Power is P_t , the power flux density S (in W/m^2) at distance d is

$$S = \frac{P_t}{4\pi d^2}$$

An antenna at distance d from the transmitter receives some of this power.

This received power (P_r) will depend on 2 factors

(1) The Power flux density (S)

(2) The effective Aperture A_e of receiving Antenna,

$$P_r = S \cdot A_e$$

So Received Power

$$P_r = S A_e$$

$$P_r = \frac{P_t}{4\pi d^2} A_e$$

The effective Aperture of receiving antenna is related to its gain & wavelength (λ)

$$A_e = \frac{G_r \lambda^2}{4\pi}$$

Now substitute these values for received power equation

$$P_r = P_t \frac{G_r \lambda^2}{4\pi d^2} \frac{1}{4\pi}$$

$$P_r = P_t G_r \times \left(\frac{\lambda}{4\pi d} \right)^2$$

If the transmitting antenna has gain G_t then equation becomes

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

Path loss can be defined as transmitted power to received power

$$\text{Path loss } (P_L) = \frac{P_t}{P_r}$$

or

$$P_L = \frac{1}{G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2}$$

Received Power

$$P_r(\text{dB}) = P_t + G_t + G_r - L(\text{dB})$$

⑧ If both Antenna are isotropic (radiate energy in all direction)

then $G_T G_R = 1$

So $P_L = \left(\frac{4\pi d^2}{\lambda^2} \right) \cdot \text{Imp.}$

Convert it into dB

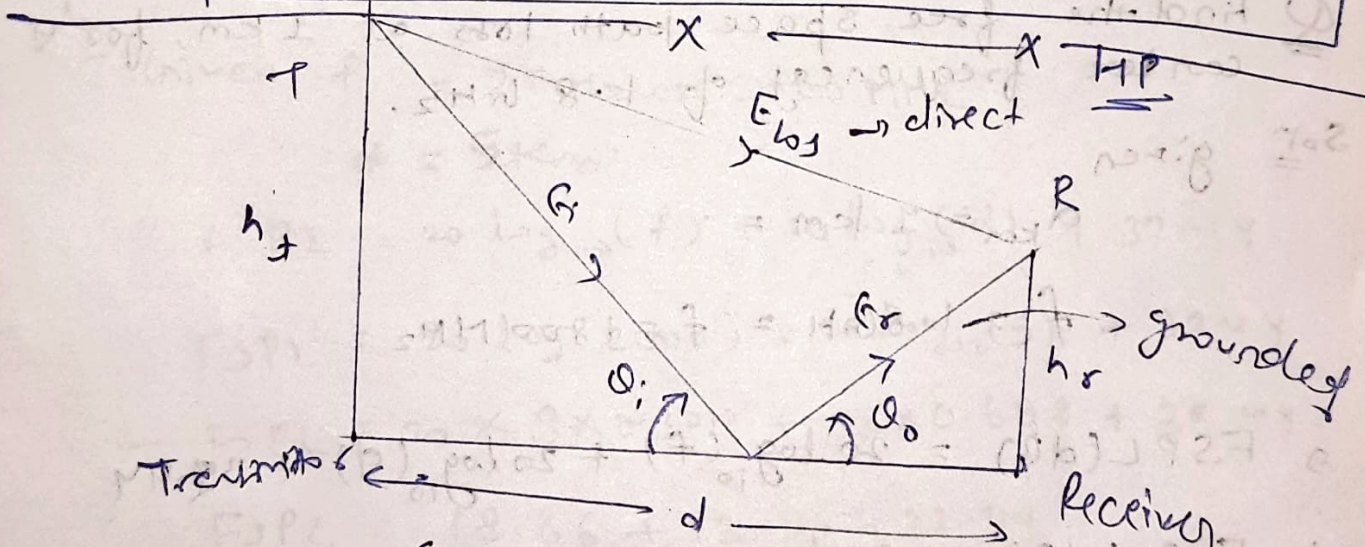
$$FSP_L (\text{dB}) = 10 \log_{10} \left(\frac{4\pi d^2}{\lambda^2} \right)$$

$$FSP_L (\text{dB}) = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) \quad \because \lambda = \frac{c}{f}$$

$c = 3 \times 10^8 \text{ m/s}$

$$FSP_L (\text{dB}) = 20 \log_{10} \left(\frac{4\pi}{c} \right) + 20 \log_{10} (df)$$

$$* FSP_L (\text{dB}) = 20 \log_{10} (d) + 20 \log_{10} (f) - 147.5 *$$



(Two Ray Ground Reflection mode)

$$E_{\text{Total}} = E_{L0} + E_{R0}$$

* Two Ray Model :-

(2)

Two Ray model in wireless communication is widely used for analyzing the propagation of radio waves in the presence of obstacles like tall buildings, trees, crowd, street light etc.

• Components of the model :-

(1) Direct Path Straight line between the transmitter & receiver.

(2) Reflective Path

Signal reflect of the ground & then reaches the receiver.

At large distance the received power P_r can be approximately calculated by $P_r(d)$

$$** \left[P_r(d) = \frac{P_t G_r G_t h_t^2 h_r^2}{d^4 L} \right]$$

Here

P_r = Power of receiver

P_t = Power of Tx

G_t = Gain of transmission Antenna

G_r = Gain of Receiver Antenna

h_t = height of Tx Antenna

h_r = height of Rx Antenna

L = Power loss (including Path & Reflection loss)

d = distance b/w Tx & Rx

$$** \left(P_r(d) \propto \frac{1}{d^4} \right)$$

$$P_r \propto \frac{1}{d^4}$$

• Cross over distance :-

$$d_c = \frac{4\pi h_t h_r}{\lambda}$$

critical distance

$$L_{dB} = 40 \log_{10}(d) - 10 \log_{10}(h_t) - 10 \log_{10}(h_r) - 20 \log_{10}(h_t) - 20 \log_{10}(h_r)$$

$$L_{dB} = 40 \log_{10}(d) - 10 \log_{10}(h_t + h_r) - 20 \log_{10}(h_t + h_r)$$

Case

$$d < d_c$$

Free space Model is more accurate

$$d > d_c$$

Two ray model is more accurate.

Application

- Cellular Networks
- Tele Communication System.
- Wifi S.I.O.T. Planning
- Rural & Suburban path loss modeling.

Q Find the free space path loss at 1 km, for a carrier frequency of 1.8 GHz.

Sol given

$$d_{km} = 1 \text{ km} =$$

$$f = 1.8 \text{ GHz} = f = 1800 \text{ MHz}$$

$$\Rightarrow \text{FSPL(dB)} = 20 \log_{10}(f) + 20 \log_{10}(d) + 32.44$$

$$\Rightarrow \text{FSPL(dB)} = 20 \log_{10}(1800) + 20 \log_{10}(1) + 32.44$$

$$\Rightarrow \text{FSPL(dB)} = 3.25 \times 20 + 0 + 32.44$$

$$\Rightarrow \text{FSPL(dB)} = 65 + 32.44$$

$$\Rightarrow \text{FSPL(dB)} = 97.44$$

Q A wireless communication system operates at a frequency of 900 MHz. The distance between Tx & Rx is 2 km. Calculate the FSPL.

Solⁿ Given $f = 900 \text{ MHz}$

$d = 2 \text{ km}$

$$\therefore \text{FSPL} = 20 \log_{10}(f) + 20 \log_{10}(d) + 32.44$$

$$\text{FSPL} = 20 \times 2.954 + 20 \times 0.3010 + 32.44$$

$$\text{FSPL} = 59.08 + 6.020 + 32.44$$

$$\boxed{\text{FSPL} = 97.54 \text{ dB}}$$

Q A wireless communication system operates at a frequency of 2 GHz. The distance between Tx & Rx is 5 km. Calculate the FSPL.

Ans Given $f = 2 \text{ GHz} = 2000 \text{ MHz}$

$d = 5 \text{ km}$

$$\therefore \text{FSPL} = 20 \log_{10}(f) + 20 \log_{10}(d) + 32.44$$

$$\text{FSPL} = 20 \log(2000) + 20 \log_{10}(5) + 32.44$$

$$\text{FSPL} = 20 \times 3 \times 0.3010 + 20 \times 0.698 + 32.44$$

$$\text{FSPL} = 18.06 + 13.96 + 32.44$$

$$\boxed{\text{FSPL} = 64.46 \text{ dB}}$$

Q. A transmitter & receiver have antenna gain of 1. Transmitter Power is 1 Watt. Height of transmitter & Receiver Antenna is respectively 50m & 2m. Separation distance is 1 km. Calculate received power using 2 Ray model.

Solⁿ Given.

Antenna gain = 1

Tx Power = 1 Watt

$H_{Rx} = 2 \text{ m}$

$H_{Tx} = 50 \text{ m}$

$d = 1 \text{ km}$

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4 L}$$

$$P_r = 1 \times 1 \times 1 \times \frac{(50)^2 (2)^2}{(1000)^4 \times 1}$$

$$P_r = \frac{25 \times 4 \times 100}{10^{12} \times 10^{10}}$$

$$P_r = \frac{100 \times 4}{10^{19}} = \left\{ P_r = 10^{-8} \text{ Watt} \right\}$$

$$P_r = 500 \times 10^{10}$$

$$P_r = 5 \times 10^{12}$$

Q. A transmitter radiates $P_t = 20\text{ W}$ & transmitter & receiver height are 40 m & 5 m distance between them is 1 km .
 assume $G_t = G_r = 1$ (isotropic)
 find Received Power.

Soln Given

$$P_t = 20\text{ W}$$

$$P_r = ?$$

$$G_t = G_r = 1$$

$$H_t = 40\text{ m}$$

$$H_r = 5\text{ m}$$

$$d = 1\text{ km} \quad L = 1$$

$$P_r = \frac{P_t G_r G_t h_t^2 h_r^2}{d^4 L}$$

$$P_r = \frac{20 \times 1 \times 1 \times (40)^2 \times (5)^2}{(1000)^4 \times 1}$$

$$P_r = \frac{2 \times 16 \times 25 \times 10^3}{10^{12}}$$

$$P_r = \frac{8 \times 10^5}{10^{12}}$$

$$P_r = 8 \times 10^{-7} \text{ watt}$$

Q In a two ray model Transmitted Power is 10 W . Received power is $P_r = 10^{-7}\text{ W}$ & $d = 1000\text{ m}$. Assume $G_r = G_t = 1$ ($\therefore L = 1$) Find height of Transmitter antenna while Receiver antenna has 2 m height.

Solⁿ

Given

$$P_t = 10\text{ W}$$

$$P_r = 10^{-7}\text{ W}$$

$$h_t = h_r = 1$$

$$d = 1000\text{ m}$$

$$P_r = \frac{G_t \times G_r \times h_r^2 \times h_t^2 \times P_t}{d^4 L}$$

$$10^{-7} = P_r = \frac{1 \times 1 \times (2\text{ m})^2 \times (h_t)^2 \times (10)}{(1000)^4 \times 1}$$

$$h_t^2 = \frac{10^{-7} \times 10^{12}}{4 \times 10^9}$$

$$h_t^2 = \frac{10^4}{4}$$

$$h_t = \frac{10^2}{2}\text{ m}$$

$$\boxed{h_t = 50\text{ m}}$$

Q Transmitter has power is 20W, Antenna gain is 10dB, Receiver Antenna gain is 5dB & Total path loss is 120dB. Find the received power in dB.

Solⁿ

Given

$$P_t = 20$$

$$P_r = ?$$

$$G_r = 5 \text{ dB}$$

$$G_t = 20 \text{ W}$$

$$P_r = \frac{G_t G_r h_r^2 h_t^2}{d^4 L} P_t \quad P_r = P_t + G_t + G_r \times \frac{1}{L}$$

$$P_r = 20 \times 5 \times 20 \times$$

$$P_r = \frac{20 \times 5 \times 20}{1}$$

$$P_r = 2000 \text{ watt}$$

for dB.

$$P_t(\text{dB}) = 10 \log_{10} (200000)$$

$$P_t(\text{dB}) = 43.01 \text{ dB}$$

$$P_r = P_t + G_t + G_r - \text{Path Loss}$$

$$P_r = -61.99 \text{ dB}$$

* Link Budget Design

→ It is the accounting of all gains & losses in a communication link to determine the received signal power.

$$P_r = P_t + G_t + G_r - L$$

P_t = transmitted power

L = path loss

G_t = transmitter antenna gain

P_r = Received power

G_r = receiver antenna gain

~~Not~~ $L = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right)$

■ Small Scale fading

It is the variation in the received signal strength with time / position due to changes in the wireless propagation channel.

→ Causes of fading

- 1) multipath propagation
- 2) Doppler shift
- 3) Movement of obstacles
- 4) changes in the propagation environment

There are two types of Fading.

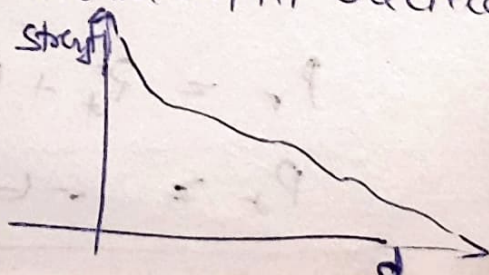
- 1) Large scale fading
- 2) Small scale fading.

→ Large Scale fading → long-term variation in received signal strength.

→ Large distance.

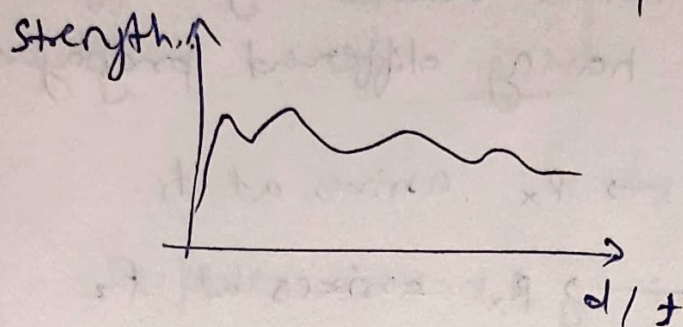
Causes → path loss, Shadowing

& long environment change.

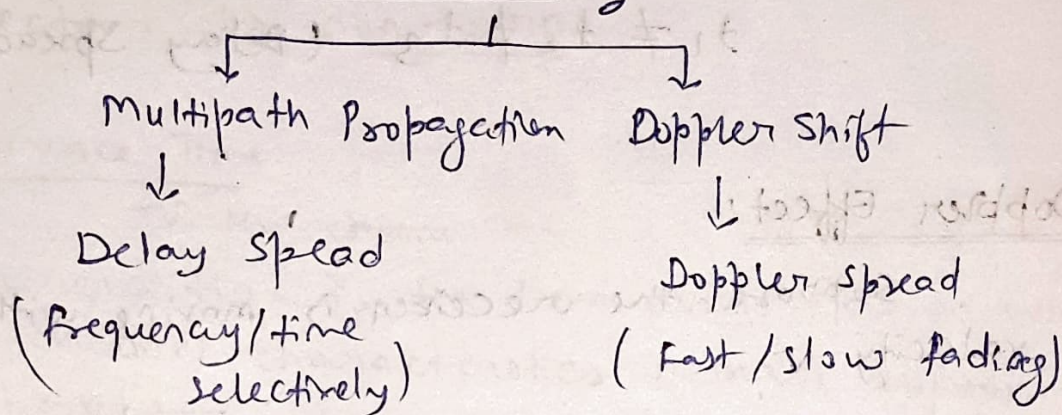


2) Small scale fading

Rapid variation in received signal strength over a short distance / short time



* Causes of small scale fading



1) Multipath Propagation

A transmitted signal may reach the receiver through different path due to reflection, diffraction & scattering.

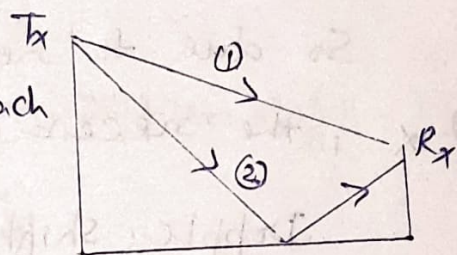
→ It gets multiple copies of same signal.

So original signal = A_0

Reflected = A_1

Scattered = A_3

all are different



Constructive interference = $A+A = 2A$ (Strong signal)

Destructive interference = $A-A = 0$ (weak signal)

• Multipath - Delay spread

Delay spread describes the time dispersion of a multipath channel caused by different propagation paths having different propagation delays.

path 1 $\rightarrow$ R_x arrives at t_1

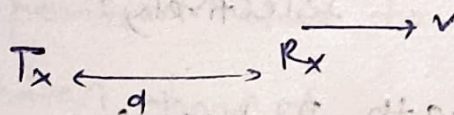
path 2 $\rightarrow$ R_x arrives at t_2

path 3 $\rightarrow$ R_x arrives at t_3

$t_1 \neq t_2 \neq t_3$ (Delay Spread)

2) Doppler Effect:-

Suppose the receiver is moving with velocity v



So due to relative motion between Tx & Rx, the received frequency changes.

Doppler shift

$$f_d = \frac{v}{\lambda} \cos \theta$$

Here

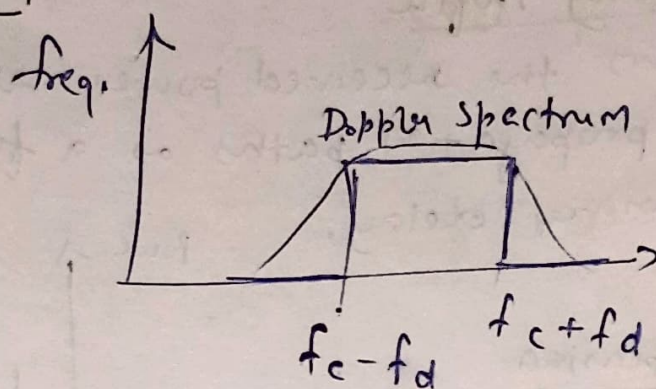
f_d = Doppler shift
in Hz

v = velocity of Rx in

m/s

θ = angle of arrival
signal.

* Doppler Spread



Doppler spread B_D is the range of frequencies over which the received doppler spectrum is non-zero

* Coherence Time

~~If the channel~~

Coherence time is the time duration over which the channel characteristics remain approximately unchanged.

$$(T_c \approx \frac{1}{f_m})$$

f_m is maximum Doppler shift.

$$\text{So } f_m \uparrow \rightarrow T_c \downarrow$$

means more doppler $\rightarrow$ channel changes faster & smaller coherence time.

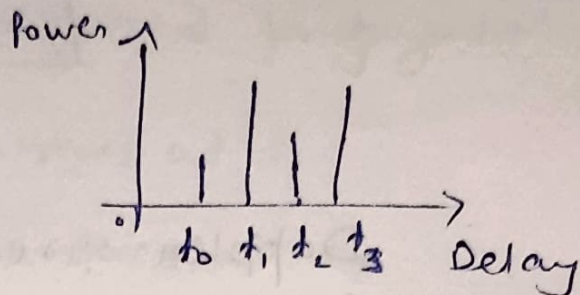
Parameters of Multipath channel

different copies of the transmitted signal arrive at the receiver with different delays & power.

Received Power = Multiple delayed Copies

• Power Delay Profile

It shows the received power associated with different propagation paths as a function of their arrival delay.



Time Dispersion

Because different waves arrive at different times.

$T_1 \neq T_2 \neq T_3$ spread in time at receiver.

• Mean Access Delay

average delay of multiple path components relative to first arriving path, weighted according to their received power.

$$\bar{T} = \frac{\sum_i P_i T_i}{\sum_i P_i}$$

P_i = power of i th path

T_i = delay at i th path

$\bar{T}$ = Mean Access delay.

• RMS Delay Spread

$$\sigma = \sqrt{\text{MAD}}$$

$$\sigma = \sqrt{\frac{\sum_i P_i T_i^2}{\sum_i P_i} - \bar{T}^2}$$

ISI (Inter-Symbol Interference)

large delay spread then more time dispersion
So greater possibility of ISI.

Coherence Bandwidth

$$B_c = \frac{1}{5\sigma_T}$$

So $\sigma_T \uparrow B_c \downarrow$

So Multipath fading

↓
flat fading

$$B_s < B_c$$

↓
freq. selective fading

$$B_s > B_c$$

B_s = Bandwidth of Signal

B_c = Coherence Bandwidth of channel.

Flat fading Occurs when
signal bandwidth B_s
smaller than B_c

→ Nearly constant channel
response

→ Delay spread small

→ Generally negligible ISI
from multipath

freq. selective fading occurs
when signal bandwidth B_s
greater than B_c .

Varies with frequency.

large delay spread.

Significant possibility of
ISI

* Fading due to doppler effect

Doppler shift =

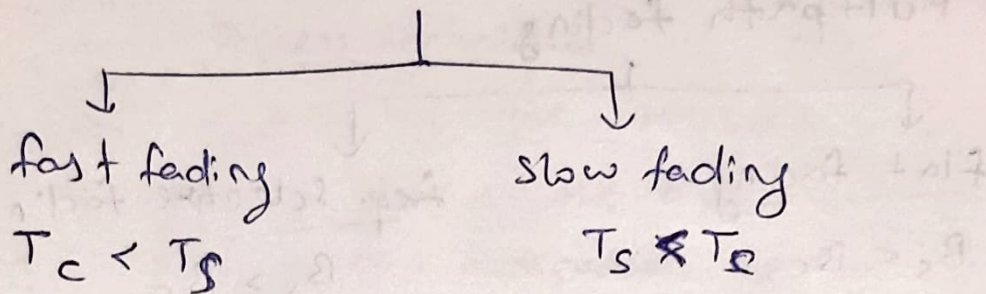
$$f_d = \frac{v}{\lambda} \cos \theta$$

$$\text{maximum } f_d = \frac{v}{\lambda} \Rightarrow \cos \theta = 1$$

or

$$f_m = \frac{v f_c}{c}$$

$f_c = \text{freq. carrier}$



Here $T_c = \text{Coherence time}$

$T_s = \text{symbol duration}$

+ Channel variation fast	+ Channel variation slow
+ small coherence time	+ large coherence time
+ Relatively large doppler	+ Relatively small doppler
+ Occurs when channel changes rapidly compared with the duration of the transmitted signal/symbol	+ Occurs when channel changes significantly slowly compared with the signal duration